

1. Conceptual Data Model

- Entities:
- Customer
- A party that has made at least one order and is tracked for order and spend analytics.
- Order
- A purchase transaction for one item by a customer.
- OrderSummary
- Historical record combining customer and order data; supports change tracking with SCD Type 2 logic.
- CustomerAggregateSpend
- Daily aggregated spend totals per customer.
- Relationships:
- Each Order is linked to one Customer (cust_id).
- Each OrderSummary is produced by joining Customer and Order (on cust_id), with SCD Type 2 logic.
- Each CustomerAggregateSpend is derived by grouping OrderSummary by name and date.

2. Logical Data Model

- Customer
- cust_id [Primary Key]
- name
- email_id
- region
- Order
- order_id [Primary Key]
- item_name
- price_per_unit
- qty
- date
- cust_id [Foreign Key → Customer.cust_id]
- OrderSummary
- cust_id [Foreign Key → Customer.cust_id]
- name
- email_id

- region
- order_id
- item_name
- price_per_unit
- qty
- date
- start_date [for SCD Type 2 handling]
- end_date [for SCD Type 2 handling]
- active [for SCD Type 2 handling]
- CustomerAggregateSpend
- name
- total_amount
- date
- Cardinality:
 - Customer (1) — (0..N) Order
 - Customer (1) — (0..N) OrderSummary
 - Order (1) — (1) OrderSummary
 - OrderSummary (0..N) — (1) CustomerAggregateSpend
- Normalization:
 - All tables are normalized for atomicity (1NF), full non-key dependency (2NF), and no transitive dependency (3NF), except for intentional denormalization in aggregation.
- Business Rules:
 - TotalAmount = price_per_unit × qty
 - Remove "Null"/Null literals from base tables
 - Remove duplicate records from base tables
 - SCD Type 2 tracking in OrderSummary

3. Physical Data Model

- Table: customer
 - cust_id: STRING, PRIMARY KEY, NOT NULL
 - name: STRING, NOT NULL
 - email_id: STRING, NULLABLE
 - region: STRING, NULLABLE
- Table: order

- order_id: STRING, PRIMARY KEY, NOT NULL
 - item_name: STRING, NOT NULL
 - price_per_unit: DECIMAL(18,2), NOT NULL
 - qty: INTEGER, NOT NULL
 - date: DATE, NOT NULL
 - cust_id: STRING, FOREIGN KEY, NOT NULL
 - total_amount: DECIMAL(18,2), NOT NULL (computed)
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- Table: ordersummary
 - cust_id: STRING, FOREIGN KEY, NOT NULL
 - name: STRING, NOT NULL
 - email_id: STRING, NULLABLE
 - region: STRING, NULLABLE
 - order_id: STRING, NOT NULL
 - item_name: STRING, NOT NULL
 - price_per_unit: DECIMAL(18,2), NOT NULL
 - qty: INTEGER, NOT NULL
 - date: DATE, NOT NULL
 - start_date: DATE, NULLABLE
 - end_date: DATE, NULLABLE
 - active: BOOLEAN, NOT NULL
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- Table: customeraggregatespend
 - name: STRING, NOT NULL
 - total_amount: DECIMAL(18,2), NOT NULL
 - date: DATE, NOT NULL
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- Indexing & Constraints:
 - All PKs indexed
 - FK: order.cust_id → customer.cust_id
 - FK: ordersummary.cust_id → customer.cust_id
 - Composite index/partition by date for large tables

Naming Conventions

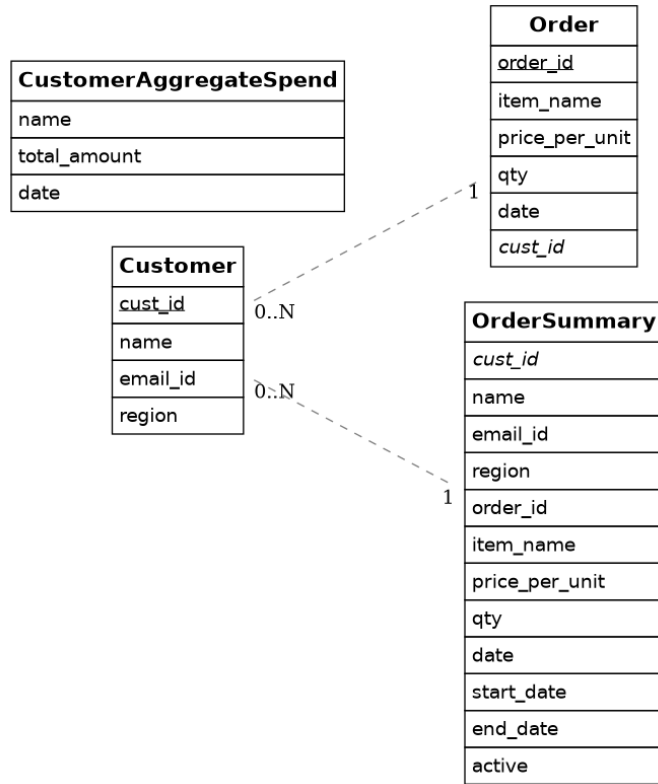
- snake_case for tables and columns
- PKs: suffixed with _id
- Foreign keys: prefixed with +

- Computed/derived: explicitly named (total_amount)
- SCD tracking: start_date, end_date, active

Data Dictionary

Entity	Attribute	Definition	Type	Notes
customer	cust_id	Unique customer identifier	STRING	PK
customer	name	Customer name	STRING	
customer	email_id	Customer email	STRING	Nullable
customer	region	Sales region	STRING	Nullable
order	order_id	Order identifier	STRING	PK
order	item_name	Name of item ordered	STRING	
order	price_per_unit	Unit price	DECIMAL(18,2)	
order	qty	Quantity ordered	INTEGER	
order	date	Order date	DATE	
order	cust_id	Customer reference	STRING	FK to customer
order	total_amount	price_per_unit × qty	DECIMAL(18,2)	Computed
ordersummary	cust_id	Customer reference	STRING	FK to customer
ordersummary	name	Customer name	STRING	
ordersummary	email_id	Email address	STRING	Nullable
ordersummary	region	Sales region	STRING	Nullable
ordersummary	order_id	Order identifier	STRING	
ordersummary	item_name	Name of item	STRING	
ordersummary	price_per_unit	Unit price	DECIMAL(18,2)	
ordersummary	qty	Quantity	INTEGER	
ordersummary	date	Date	DATE	
ordersummary	start_date	SCD2: Start date of record	DATE	
ordersummary	end_date	SCD2: End date of record	DATE	
ordersummary	active	SCD2: Indicates active record	BOOLEAN	
customeraggregatespend	name	Customer name	STRING	Grouped
customeraggregatespend	total_amount	Aggregated spend	DECIMAL(18,2)	SUM of TotalAmount
customeraggregatespend	date	Spend date	DATE	Grouped

Text-Based ERD



Order 1--1 OrderSummary

OrderSummary *--1 CustomerAggregateSpend